

Q2N.

1. A delivery system needs optimal routing.
Create pseudocode to model the problem as a graph and compute efficient delivery routes using shortest path or MST.

PSEUDOCODE:

STEP 1: Creating the graph

INPUT number of delivery locations V

INPUT number of roads E

Create an empty graph G

FOR each road

INPUT source, destination, distance

ADD edge (u, v) to G

RETURN G

END FOR

FUNCTION Dijkstra (Graph G , Source depot):

FOR each vertex v IN $G.V$:

~~dist~~ $[v] \leftarrow \text{INFINITY}$

~~prev~~ $[v] \leftarrow \text{NULL}$

$\text{dist}[S] \leftarrow 0$

CREATE empty priority queue Q

INSERT $(0, S)$ into Q

WHILE Q is not empty

Remove vertex U with minimum distance

FOR each adjacent vertex V of U

$newDistance \leftarrow distance[U] + weight(U, V)$

IF $newDistance < distance[V]$ THEN

$distance[V] \leftarrow newDistance$

$predecessor[V] \leftarrow U$

Insert $(distance[V], V)$ into Q

END IF

END FOR

END WHILE

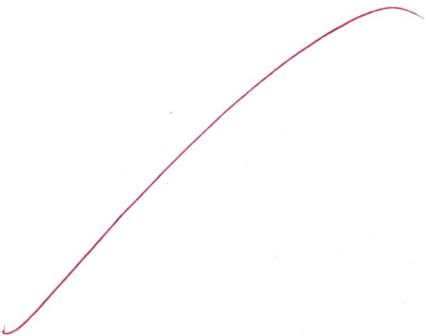
FOR each destination vertex

Display shortest distance from S

Display the route using $predecessor[]$

END FOR

END



2. Write pseudocode for constructing a graph in MST using Kruskal's Algorithm.

A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

1. All zones must be connected
2. Total installation cost must be minimised.
3. No redundant connections (no-cycles) allowed.
4. The network must remain fully connected.

PSEUDOCODE:

INPUT number of vertices V

INPUT number of edges E

INPUT all edges with:

Source

Destination

Cost

Sort all edges in increasing order

Create a separate set for each vertex

$MST \leftarrow$ empty set

$TotalCost \leftarrow 0$

FOR each edge $(u, v, cost)$ in sorted order

IF $Find(u) \neq Find(v)$ THEN

ADD edge $(u, v, cost)$

TotalCost $\leftarrow$ TotalCost + Cost

Union (u, v)

END IF

IF number of edges in MST = $V-1$ THEN

BREAK

END IF

END FOR

IF number of edges in MST = $V-1$ THEN

Display "Minimum Spanning tree"

Display all edges in MST

Display "Total installation cost:", TotalCost

ELSE

Display "Error"

END IF

END

